

MH3100 Real Analysis I

Tutorial 2 Solutions

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Overview

- **Lecture coverage:** Lectures 4–6 — the definition of the limit of a sequence, the Algebraic Limit Theorem, the Order Limit Theorem, the Squeeze Theorem, and Cauchy sequences.
- **Questions:** 8 questions from Tutorial Question Set No. 2 (main Q1–Q5, Additional Questions A1–A3); solutions extend the official Tutorial 2 hints.
- **Notation:** as in lectures — $\mathbb{N} = \{1, 2, 3, \dots\}$ (does *not* include 0); a sequence is $\{a_n\}$; a limit is written value-first, $a = \lim a_n$; a property holding eventually means it holds on some tail $a_N, a_{N+1}, \dots$.
- **Running restriction:** Q4 and Q5 explicitly forbid the Algebraic Limit Theorem and (for Q5) the Cauchy Criterion; every proof in those two questions is therefore built from the raw ε - N definitions only.

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Question 1 (Eight true/false statements about convergence and divergence)

Problem

Determine if each of the following statements is true.

- (a) There is a sequence with an infinite number of 1's and does not converge to 1.
- (b) There is a sequence with an infinite number of 1's and converges to a limit not equal to 1.
- (c) There is a divergent sequence such that for every $n \in \mathbb{N}$ it is possible to find n consecutive ones somewhere in the sequence.
- (d) There are sequences $\{a_n\}$ and $\{b_n\}$, which both diverge, but whose sum $\{a_n + b_n\}$ converges.
- (e) There are sequences $\{a_n\}$ and $\{b_n\}$, where $\{a_n\}$ converges, $\{b_n\}$ diverges, and $\{a_n + b_n\}$ converges.
- (f) There is a convergent sequence $\{b_n\}$ with $b_n \neq 0$ for each $n \in \mathbb{N}$ such that $\{1/b_n\}$ diverges.
- (g) There is an unbounded sequence $\{a_n\}$ and a convergent sequence $\{b_n\}$ with $\{a_n - b_n\}$ bounded.
- (h) There are sequences $\{a_n\}$ and $\{b_n\}$, where $\{a_n b_n\}$ and $\{a_n\}$ converge but $\{b_n\}$ does not.

(Tutorial 2, Q1.)

Solution

Official hint. The hints page gives only the eight answers, in order: True, False, True, True, False, True, False, True, with no justification.

Beyond the hints. Every True item below gets an explicit, verified example, and every False item gets a full proof from the Lecture 4–6 definitions (limit, boundedness, Algebraic Limit Theorem). Part (e) is also sample midterm Q1(a).

Method 1: item-by-item construction and proof

(a) True. *Example.* Let $a_n = 1$ if n is odd and $a_n = 0$ if n is even. Then $a_n = 1$ for infinitely many $n \in \mathbb{N}$ (all odd n). *Verification that $\{a_n\}$ does not converge to 1.* Take $\varepsilon = \frac{1}{2} > 0$. For every $N \in \mathbb{N}$, the even number $n = 2 \lceil N/2 \rceil$ satisfies $n \geq N$ and $a_n = 0$, so $|a_n - 1| = 1 \not< \frac{1}{2}$. Hence there is no N for which $n \geq N$ implies $|a_n - 1| < \frac{1}{2}$, so $\{a_n\}$ eventually within $\frac{1}{2}$ of 1 fails; therefore $\{a_n\}$ does not converge to 1.

True; $a_n = 1$ (n odd), $a_n = 0$ (n even).

(b) False.

Idea. (*Scratch work – not part of the written answer.*) If a sequence converges to some $a \neq 1$, and 1 occurs infinitely often, then infinitely many terms equal to 1 must all eventually sit within a tiny ball around a – but $1 \notin$ that ball once the ball is small enough. Use $\varepsilon = |a - 1|/2$.

Proof. Suppose, for contradiction, that $\{a_n\}$ has infinitely many terms equal to 1 and converges to some $a \in \mathbb{R}$ with $a \neq 1$. Let $\varepsilon := \frac{|a - 1|}{2} > 0$. By the definition of $a = \lim a_n$, there is $N \in \mathbb{N}$ such that for all $n \geq N$, $|a_n - a| < \varepsilon$. Since $a_n = 1$ occurs for infinitely many $n \in \mathbb{N}$, there is some index $n_0 \geq N$ with $a_{n_0} = 1$. Then $|1 - a| < \varepsilon = \frac{|a - 1|}{2}$, which rearranges to $|a - 1| < \frac{|a - 1|}{2}$, i.e. $\frac{|a - 1|}{2} < 0$. Since $|a - 1| > 0$ (as $a \neq 1$), this is a contradiction. Therefore no such convergent sequence with limit $a \neq 1$ exists; a sequence with infinitely many 1's can only converge to 1 (or diverge). $\square$

False; a sequence with infinitely many 1's can only converge to 1, never to another limit.

(c) **True.** *Example.* Build $\{a_n\}$ from blocks: a run of k consecutive 1's followed by a single 0, for $k = 1, 2, 3, \dots$ in turn:

$$\underbrace{1}_{k=1}, 0, \underbrace{1, 1}_{k=2}, 0, \underbrace{1, 1, 1}_{k=3}, 0, \underbrace{1, 1, 1, 1}_{k=4}, 0, \dots$$

For every $n \in \mathbb{N}$, the block of length $k = n$ supplies n consecutive 1's, so the required property holds. Every block ends in a single 0, so $a_n = 0$ occurs for infinitely many n , and every block contains a 1, so $a_n = 1$ also occurs for infinitely many n . *Verification that $\{a_n\}$ diverges.* Suppose for contradiction $a = \lim a_n$ for some $a \in \mathbb{R}$. Take $\varepsilon = \frac{1}{2} > 0$; by the definition of limit, there is $N \in \mathbb{N}$ such that $n \geq N \implies |a_n - a| < \frac{1}{2}$. Since 0's and 1's each occur infinitely often, choose $n_1, n_2 \geq N$ with $a_{n_1} = 0$ and $a_{n_2} = 1$. Then $|a_{n_1} - a| < \frac{1}{2}$ and $|a_{n_2} - a| < \frac{1}{2}$, so by the triangle inequality

$$1 = |1 - 0| = |a_{n_2} - a_{n_1}| \leq |a_{n_2} - a| + |a - a_{n_1}| < \frac{1}{2} + \frac{1}{2} = 1,$$

i.e. $1 < 1$, a contradiction. Hence $\{a_n\}$ diverges.

True; growing runs of 1's (length n) separated by single 0's, diverging by the argument above.

(d) **True.** *Example.* Let $a_n = n$ and $b_n = -n$ for all $n \in \mathbb{N}$. *Verification that $\{a_n\}$ and $\{b_n\}$ both diverge.* By the Theorem (Boundedness of Convergent Sequences), every convergent sequence is bounded. The sequence $\{a_n\} = \{n\}$ is unbounded: for any proposed bound $M > 0$, the Archimedean Property gives some $n \in \mathbb{N}$ with $n > M$, so $|a_n| = n > M$. Hence $\{a_n\}$ is not bounded, so it cannot be convergent; it diverges. The same argument (with $|b_n| = n$) shows $\{b_n\} = \{-n\}$ diverges. *Verification that $\{a_n + b_n\}$ converges.* $a_n + b_n = n + (-n) = 0$ for every $n \in \mathbb{N}$, so $\{a_n + b_n\}$ is the constant sequence 0, which converges to 0.

True; $a_n = n$, $b_n = -n$, $a_n + b_n = 0$ for all n .

(e) **False.** (This part is also sample midterm Q1(a).)

Idea. (*Scratch work – not part of the written answer.*) If $\{a_n\}$ and $\{a_n + b_n\}$ both converge, write $b_n = (a_n + b_n) - a_n$ and apply the Algebraic Limit Theorem (difference of two convergent sequences) to force $\{b_n\}$ to converge, contradicting the hypothesis that it diverges.

Proof. Suppose, for contradiction, that $\{a_n\}$ converges to $a \in \mathbb{R}$, $\{b_n\}$ diverges, and $\{a_n + b_n\}$ converges to some $s \in \mathbb{R}$. For every $n \in \mathbb{N}$, $b_n = (a_n + b_n) + (-1)a_n$. Since $\{a_n + b_n\}$ converges to s and $\{a_n\}$ converges to a , the Algebraic Limit Theorem (parts 1 and 2, scalar multiple and sum) implies that $\{(a_n + b_n) + (-1)a_n\} = \{b_n\}$ converges to $s + (-1)a = s - a$. This contradicts the hypothesis that $\{b_n\}$ diverges. Therefore no such triple of sequences exists. $\square$

False; if $\{a_n\}$ and $\{a_n + b_n\}$ converge then $\{b_n\} = \{(a_n + b_n) - a_n\}$ converges, by the Algebraic Limit Theorem.

(f) True. *Example.* Let $b_n = 1/n$ for each $n \in \mathbb{N}$; then $b_n \neq 0$ for all n , and $\{b_n\}$ converges to 0 (given $\varepsilon > 0$, the Archimedean Property gives $N \in \mathbb{N}$ with $N > 1/\varepsilon$, so $n \geq N$ implies $|1/n - 0| = 1/n \leq 1/N < \varepsilon$). *Verification that $\{1/b_n\}$ diverges.* $1/b_n = n$ for every n . By the Theorem (Boundedness of Convergent Sequences), a convergent sequence is bounded, but $\{n\}$ is unbounded (Archimedean Property, as in part (d)). Hence $\{1/b_n\} = \{n\}$ diverges.

True; $b_n = 1/n \rightarrow 0$, $b_n \neq 0$, but $1/b_n = n$ diverges (unbounded).

(g) False.

Idea. (*Scratch work – not part of the written answer.*) If $\{a_n - b_n\}$ is bounded and $\{b_n\}$ converges (hence is bounded, by Boundedness of Convergent Sequences), then $a_n = (a_n - b_n) + b_n$ is a sum of two bounded sequences, hence bounded, by the triangle inequality – contradicting the hypothesis that $\{a_n\}$ is unbounded.

Proof. Suppose, for contradiction, that $\{a_n\}$ is unbounded, $\{b_n\}$ converges, and $\{a_n - b_n\}$ is bounded. Since $\{a_n - b_n\}$ is bounded, there is $M_1 > 0$ with $|a_n - b_n| \leq M_1$ for all $n \in \mathbb{N}$. Since $\{b_n\}$ converges, the Theorem (Boundedness of Convergent Sequences) gives $M_2 > 0$ with $|b_n| \leq M_2$ for all $n \in \mathbb{N}$. Then for every $n \in \mathbb{N}$, the triangle inequality gives

$$|a_n| = |(a_n - b_n) + b_n| \leq |a_n - b_n| + |b_n| \leq M_1 + M_2.$$

Therefore $\{a_n\}$ is bounded (by $M_1 + M_2 > 0$), contradicting the hypothesis that $\{a_n\}$ is unbounded. Hence no such pair of sequences exists. $\square$

False; $a_n = (a_n - b_n) + b_n$ is bounded whenever $\{a_n - b_n\}$ and $\{b_n\}$ are, by the triangle inequality.

(h) True. *Example.* Let $a_n = 0$ for all $n \in \mathbb{N}$ (so $\{a_n\}$ converges to 0), and let $b_n = (-1)^n$ for all $n \in \mathbb{N}$. *Verification that $\{b_n\}$ diverges.* Suppose $b = \lim b_n$ for some $b \in \mathbb{R}$. Take $\varepsilon = 1 > 0$; there is $N \in \mathbb{N}$ with $n \geq N \implies |b_n - b| < 1$. Choose an even $n_1 \geq N$ and an odd $n_2 \geq N$; then $b_{n_1} = 1$ and $b_{n_2} = -1$, so $|1 - b| < 1$ and $|-1 - b| < 1$. By the triangle inequality, $2 = |1 - (-1)| \leq |1 - b| + |b - (-1)| < 1 + 1 = 2$, i.e. $2 < 2$, a contradiction. Hence $\{b_n\}$ diverges. *Verification that $\{a_n b_n\}$ converges.* $a_n b_n = 0 \cdot (-1)^n = 0$ for every n , so $\{a_n b_n\}$ is the constant sequence 0, which converges to 0.

True; $a_n = 0 (\rightarrow 0)$, $b_n = (-1)^n$ (diverges), $a_n b_n = 0 (\rightarrow 0)$.

Common slip. In parts (b), (e), (g) it is tempting to just assert the contradiction ("but that's impossible") without exhibiting the specific ε that breaks the definition, or without invoking a named theorem (Algebraic Limit Theorem in (e); Boundedness of Convergent Sequences plus the triangle inequality in (g)). On a timed paper this loses marks even when the final True/False answer is correct, since Q1/Q2-style questions on the actual midterm are graded on the answer alone but Q3/Q4-style "prove" questions are graded on the argument.

Question 2 (ε - N verification of three limits)

Problem

Verify, using the definition of limits of sequences, that the following sequences converge to the proposed limits.

$$(a) \quad \frac{2}{5} = \lim_{n \rightarrow \infty} \frac{2n+1}{5n+4}.$$

$$(b) \quad 0 = \lim_{n \rightarrow \infty} \frac{2n^2}{n^3+3}.$$

$$(c) \quad 0 = \lim_{n \rightarrow \infty} \frac{\sin(n^2)}{\sqrt[3]{n}}.$$

(Tutorial 2, Q2.)

Solution

Official hint. (i) Use the Archimedean Property to find an appropriate tail sequence for each of the three limits. (ii) For (c), recall that $-1 \leq \sin(x) \leq 1$ for any real number x .

Beyond the hints. The hints do not carry out any of the three bounding computations or state the resulting choice of $N(\varepsilon)$. All three are done in full below; each needs a different algebraic bound.

Method 1: bound $|a_n - a|$ by C/n^p , then invert for N

$$(a) \quad \frac{2}{5} = \lim_{n \rightarrow \infty} \frac{2n+1}{5n+4}.$$

Idea. (*Scratch work – not part of the written answer.*) Combine the fraction over a common denominator and simplify the numerator; the leading terms cancel, leaving a constant over $5n+4$, which is easy to bound by a constant over n .

$$\left| \frac{2n+1}{5n+4} - \frac{2}{5} \right| = \left| \frac{5(2n+1) - 2(5n+4)}{5(5n+4)} \right| = \left| \frac{10n+5-10n-8}{5(5n+4)} \right| = \frac{3}{5(5n+4)} < \frac{3}{25n}$$

(using $5n+4 > 5n$ for $n \geq 1$). So it suffices to make $3/(25n) < \varepsilon$, i.e. $n > 3/(25\varepsilon)$.

Proof. Let $\varepsilon > 0$ be given. By the Archimedean Property, choose $N \in \mathbb{N}$ with $N > \frac{3}{25\varepsilon}$. Then for all $n \geq N$,

$$\left| \frac{2n+1}{5n+4} - \frac{2}{5} \right| = \frac{3}{5(5n+4)} < \frac{3}{25n} \leq \frac{3}{25N} < \varepsilon.$$

Therefore, by the definition of the limit of a sequence, $\frac{2}{5} = \lim_{n \rightarrow \infty} \frac{2n+1}{5n+4}$. □

$$(b) \quad 0 = \lim_{n \rightarrow \infty} \frac{2n^2}{n^3+3}.$$

Idea. (*Scratch work – not part of the written answer.*) Drop the $+3$ in the denominator to get a clean bound $2/n$.

$$\left| \frac{2n^2}{n^3 + 3} \right| = \frac{2n^2}{n^3 + 3} < \frac{2n^2}{n^3} = \frac{2}{n}.$$

So it suffices to make $2/n < \varepsilon$, i.e. $n > 2/\varepsilon$.

Proof. Let $\varepsilon > 0$ be given. By the Archimedean Property, choose $N \in \mathbb{N}$ with $N > \frac{2}{\varepsilon}$. Then for all $n \geq N$, since $n^3 + 3 > n^3 > 0$,

$$\left| \frac{2n^2}{n^3 + 3} - 0 \right| = \frac{2n^2}{n^3 + 3} < \frac{2n^2}{n^3} = \frac{2}{n} \leq \frac{2}{N} < \varepsilon.$$

Therefore, by the definition of the limit of a sequence, $0 = \lim_{n \rightarrow \infty} \frac{2n^2}{n^3 + 3}$. □

(c) $0 = \lim_{n \rightarrow \infty} \frac{\sin(n^2)}{\sqrt[3]{n}}.$

Idea. (*Scratch work – not part of the written answer.*) Since $-1 \leq \sin(x) \leq 1$ for every real x , the numerator is bounded regardless of what n^2 is; only the denominator $\sqrt[3]{n}$ needs to be made large. This gives a bound of the shape $C/n^{1/3}$ rather than C/n , so the Archimedean Property must be applied to $1/\varepsilon^3$, not $1/\varepsilon$.

$$\left| \frac{\sin(n^2)}{\sqrt[3]{n}} \right| = \frac{|\sin(n^2)|}{\sqrt[3]{n}} \leq \frac{1}{\sqrt[3]{n}}.$$

Requiring $1/\sqrt[3]{n} < \varepsilon$ is equivalent to $\sqrt[3]{n} > 1/\varepsilon$, i.e. (cubing both positive sides) $n > 1/\varepsilon^3$.

Proof. Let $\varepsilon > 0$ be given. By the Archimedean Property, choose $N \in \mathbb{N}$ with $N > \frac{1}{\varepsilon^3}$. Then for all $n \geq N$, since $-1 \leq \sin(n^2) \leq 1$ implies $|\sin(n^2)| \leq 1$,

$$\left| \frac{\sin(n^2)}{\sqrt[3]{n}} - 0 \right| = \frac{|\sin(n^2)|}{\sqrt[3]{n}} \leq \frac{1}{\sqrt[3]{n}} \leq \frac{1}{\sqrt[3]{N}} < \varepsilon,$$

where the last step follows because $N > 1/\varepsilon^3$ implies $\sqrt[3]{N} > 1/\varepsilon$, hence $1/\sqrt[3]{N} < \varepsilon$.

Therefore, by the definition of the limit of a sequence, $0 = \lim_{n \rightarrow \infty} \frac{\sin(n^2)}{\sqrt[3]{n}}$. □

Common slip. Bounding $|a_n - a|$ by an expression that still depends on n in the wrong direction (e.g. leaving $5n + 4$ in the denominator and then trying to solve for N from a quadratic-looking inequality) makes the algebra unnecessarily hard. The trick in every part is to *enlarge* the bound to something invertible (C/n or $C/n^{1/3}$) before solving for N ; the resulting N need not be optimal, only sufficient.

Question 3 (Bounded times null, and a warning when the limit is nonzero)

Problem

In this question, $\{a_n\}$ is a bounded sequence and $\{b_n\}$ is a convergent sequence.

- (a) Prove that if $\{b_n\}$ has limit 0, then the sequence $\{a_n b_n\}$ has limit 0.
- (b) Give an example to show that the sequence $\{a_n b_n\}$ may not have a limit if $\{b_n\}$ has a limit different from 0.

(Tutorial 2, Q3.)

Solution

Official hint. (a) Use the Squeeze Theorem. (b) Recall that bounded sequences may not be convergent.

Beyond the hints. The hint for (a) does not show the actual squeeze $-M|b_n| \leq a_n b_n \leq M|b_n|$ nor that the two outer sequences tend to 0; the hint for (b) supplies no concrete counterexample.

Method 1: (a) via the Squeeze Theorem; (b) via an explicit oscillating example

(a).

Idea. (Scratch work – not part of the written answer.) $|a_n| \leq M$ gives $|a_n b_n| = |a_n| |b_n| \leq M |b_n|$, i.e. $-M |b_n| \leq a_n b_n \leq M |b_n|$. If both outer sequences $\pm M |b_n|$ converge to 0, the Squeeze Theorem finishes the job. Since $b_n \rightarrow 0$ means precisely that $|b_n - 0| < \varepsilon$ eventually, this is exactly the statement $|b_n| \rightarrow 0$, so $M |b_n| \rightarrow 0$ by the Algebraic Limit Theorem (scalar multiple).

Proof. Since $\{a_n\}$ is bounded, there is $M > 0$ such that $|a_n| \leq M$ for every $n \in \mathbb{N}$. For every $n \in \mathbb{N}$,

$$|a_n b_n| = |a_n| |b_n| \leq M |b_n|, \quad \text{so that} \quad -M |b_n| \leq a_n b_n \leq M |b_n|.$$

Since $0 = \lim b_n$, the definition of the limit gives, for every $\varepsilon > 0$, some $N \in \mathbb{N}$ such that $n \geq N$ implies $|b_n - 0| = |b_n| < \varepsilon$; that is, $0 = \lim |b_n|$. By the Algebraic Limit Theorem (scalar multiple, applied twice with constants M and $-M$), the sequences $\{M |b_n|\}$ and $\{-M |b_n|\}$ both converge to $M \cdot 0 = 0$. Since $\{a_n b_n\}$ satisfies $-M |b_n| \leq a_n b_n \leq M |b_n|$ for every $n \in \mathbb{N}$ (in particular, eventually), and the two bounding sequences share the limit 0, therefore, by the Squeeze Theorem, $\{a_n b_n\}$ converges and $0 = \lim a_n b_n$. $\square$

$0 = \lim a_n b_n \text{ whenever } \{a_n\} \text{ is bounded and } 0 = \lim b_n.$

(b). *Example.* Let $a_n = (-1)^n$ (bounded, with $|a_n| \leq 1$ for all n) and let $b_n = 1$ for all $n \in \mathbb{N}$ (a constant sequence, which converges to $1 \neq 0$). *Verification that $\{a_n b_n\}$ has no limit.* $a_n b_n = (-1)^n \cdot 1 = (-1)^n$. Suppose $L = \lim (-1)^n$ for some $L \in \mathbb{R}$. Take

$\varepsilon = 1 > 0$; there is $N \in \mathbb{N}$ such that $n \geq N$ implies $|(-1)^n - L| < 1$. Choose an even $n_1 \geq N$ and an odd $n_2 \geq N$; then $|1 - L| < 1$ and $|-1 - L| < 1$, and the triangle inequality gives $2 = |1 - (-1)| \leq |1 - L| + |L - (-1)| < 1 + 1 = 2$, a contradiction. Hence $\{a_n b_n\} = \{(-1)^n\}$ diverges.

$$a_n = (-1)^n, \quad b_n = 1 \quad (\rightarrow 1 \neq 0), \quad a_n b_n = (-1)^n \text{ diverges.}$$

Common slip. Forgetting that boundedness of $\{a_n\}$ alone is not enough to invoke the Squeeze Theorem in part (a); the middle term $a_n b_n$ must be sandwiched by two sequences with the *same* limit, which is exactly what the factor $|b_n| \rightarrow 0$ supplies. In part (b), choosing a_n to be itself convergent (e.g. $a_n = 1$) trivially fails to be a counterexample, since $\{a_n b_n\}$ would then converge; the counterexample needs $\{a_n\}$ bounded but divergent.

Question 4 (Three ε - N proofs with the Algebraic Limit Theorem banned)

Problem

In this question, $\{a_n\}$ is a sequence of positive real numbers. Use the definition of limits of sequences to prove the following statements. **Do not use the Algebraic Limit Theorem.**

(a) If $2 = \lim a_n$, then $1 = \lim \frac{2a_n - 1}{3}$.

(b) If $2 = \lim a_n$, then $\sqrt{2} = \lim \sqrt{a_n}$.

(c) If $2 = \lim a_n$, then $\frac{1}{2} = \lim \frac{1}{a_n}$.

(Tutorial 2, Q4.)

Solution

Official hint. (a) Adapt the proof of $a = \lim a_n \implies ca = \lim(ca_n)$. (b) Introduce an ε' as in the proof that $a = \lim a_n$, $b = \lim b_n \implies ab = \lim a_n b_n$; taking square roots of an ε' -interval around 2 gives an asymmetric interval about $\sqrt{2}$, and one endpoint is more restrictive; isolate one square root before squaring. (c) The same ε' idea works, but the ε' -interval around 2 must avoid 0, since reciprocals near 0 blow up, and reciprocation reverses order.

Beyond the hints. None of the three parts is carried to an explicit final $N(\varepsilon)$ on the hints page. All three are completed below; part (b) is done directly by rationalizing (using positivity of a_n , given in the question) rather than via the two-sided ε' -interval sketched in the hint, which is a shorter route to the same bound.

Method 1: direct ε - N estimates from the definition of $2 = \lim a_n$

(a) $1 = \lim \frac{2a_n - 1}{3}$.

Idea. (Scratch work – not part of the written answer.)

$$\left| \frac{2a_n - 1}{3} - 1 \right| = \left| \frac{2a_n - 1 - 3}{3} \right| = \left| \frac{2a_n - 4}{3} \right| = \frac{2}{3} |a_n - 2|.$$

This is a constant multiple of $|a_n - 2|$, so the proof mirrors the scalar-multiple case with the constant $2/3$.

Proof. Let $\varepsilon > 0$ be given. Since $2 = \lim a_n$, there is $N \in \mathbb{N}$ such that for all $n \geq N$, $|a_n - 2| < \frac{3\varepsilon}{2}$. Then for all $n \geq N$,

$$\left| \frac{2a_n - 1}{3} - 1 \right| = \frac{2}{3} |a_n - 2| < \frac{2}{3} \cdot \frac{3\varepsilon}{2} = \varepsilon.$$

Therefore, by the definition of the limit of a sequence, $1 = \lim \frac{2a_n - 1}{3}$. $\square$

$$\boxed{1 = \lim \frac{2a_n - 1}{3}}.$$

(b) $\sqrt{2} = \lim \sqrt{a_n}$.

Idea. (*Scratch work – not part of the written answer.*) Since every $a_n > 0$ (given), $\sqrt{a_n}$ is defined and positive. Rationalize:

$$\sqrt{a_n} - \sqrt{2} = \frac{(\sqrt{a_n} - \sqrt{2})(\sqrt{a_n} + \sqrt{2})}{\sqrt{a_n} + \sqrt{2}} = \frac{a_n - 2}{\sqrt{a_n} + \sqrt{2}}.$$

Because $\sqrt{a_n} > 0$, the denominator satisfies $\sqrt{a_n} + \sqrt{2} > \sqrt{2}$, so $\frac{1}{\sqrt{a_n} + \sqrt{2}} < \frac{1}{\sqrt{2}}$; hence

$|\sqrt{a_n} - \sqrt{2}| < \frac{|a_n - 2|}{\sqrt{2}}$, with *no* need for an auxiliary restriction on ε' (positivity of a_n is given outright, unlike the general Algebraic Limit Theorem proof for products, where positivity of the limit is not assumed).

Proof. Let $\varepsilon > 0$ be given. Since $2 = \lim a_n$, there is $N \in \mathbb{N}$ such that for all $n \geq N$, $|a_n - 2| < \sqrt{2}\varepsilon$. Fix $n \geq N$. Since $a_n > 0$, $\sqrt{a_n}$ is a well-defined positive real number, and

$$|\sqrt{a_n} - \sqrt{2}| = \left| \frac{a_n - 2}{\sqrt{a_n} + \sqrt{2}} \right| = \frac{|a_n - 2|}{\sqrt{a_n} + \sqrt{2}} < \frac{|a_n - 2|}{\sqrt{2}} < \frac{\sqrt{2}\varepsilon}{\sqrt{2}} = \varepsilon,$$

where the first inequality uses $\sqrt{a_n} + \sqrt{2} > \sqrt{2} > 0$. Therefore, by the definition of the limit of a sequence, $\sqrt{2} = \lim \sqrt{a_n}$. $\square$

$$\boxed{\sqrt{2} = \lim \sqrt{a_n}}.$$

(c) $\frac{1}{2} = \lim \frac{1}{a_n}$.

Idea. (*Scratch work – not part of the written answer.*) $\left| \frac{1}{a_n} - \frac{1}{2} \right| = \frac{|2 - a_n|}{2a_n}$ needs a lower bound on a_n before it can be inverted into a bound on $|a_n - 2|$; the hint's ε' -interval around 2 is exactly this: force $a_n > 1$ eventually (take $\varepsilon' = 1$), which bounds the denominator $2a_n > 2$.

Proof. Since $2 = \lim a_n$, taking $\varepsilon' = 1 > 0$ in the definition of the limit gives $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$, $|a_n - 2| < 1$, which implies $a_n > 1$ (in particular $a_n \neq 0$, so $1/a_n$ is defined). Let $\varepsilon > 0$ be given; since $2 = \lim a_n$, there is $N_2 \in \mathbb{N}$ such that for all $n \geq N_2$, $|a_n - 2| < 2\varepsilon$. Let $N := \max\{N_1, N_2\}$. Then for all $n \geq N$, using $a_n > 1$ (so $2a_n > 2$),

$$\left| \frac{1}{a_n} - \frac{1}{2} \right| = \left| \frac{2 - a_n}{2a_n} \right| = \frac{|a_n - 2|}{2a_n} < \frac{|a_n - 2|}{2} < \frac{2\varepsilon}{2} = \varepsilon.$$

Therefore, by the definition of the limit of a sequence, $\frac{1}{2} = \lim \frac{1}{a_n}$. $\square$

$$\boxed{\frac{1}{2} = \lim \frac{1}{a_n}}.$$

Common slip. In part (c), bounding $|1/a_n - 1/2| = |a_n - 2|/(2a_n)$ and then immediately dividing by a_n *without first establishing a_n is bounded away from 0* is a common error: since a_n can in principle be arbitrarily close to 0 for small n , the bound $|a_n - 2|/(2a_n) < |a_n - 2|/2$ is only valid once n is large enough that $a_n > 1$, which itself requires its own application of the definition of the limit (with $\varepsilon' = 1$) *before* the main ε -argument – exactly the two-stage structure the hint's " ε' " is pointing at.

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Question 5 (Cauchy sequences: scalar multiples and squares, without the Cauchy Criterion)

Problem

In this question, $\{a_n\}$ is a Cauchy sequence and c is a real number.

- (a) Use the definition of Cauchy sequence to show that $\{ca_n\}$ is Cauchy. **Do not use the Cauchy Criterion or the Algebraic Limit Theorem.**
- (b) Use the definition of Cauchy sequence to show that $\{a_n^2\}$ is Cauchy. **Do not use the Cauchy Criterion or the Algebraic Limit Theorem.**

[Note: A proof without using the Cauchy Criterion is applicable to sequences in $\mathbb{Q}$ (or any ordered field) for which the Cauchy Criterion does not hold.] (*Tutorial 2, Q5; part (b) is also the sample midterm Q4.*)

Solution

Official hint. (i) Write/draw out the definition of $\{a_n\}$ being Cauchy; refer to the proof of the Cauchy Criterion if needed. (ii) Follow the general scheme behind the proof of the Algebraic Limit Theorem for $\{ca_n\}$ and $\{a_nb_n\}$ respectively.

Beyond the hints. The hint gives no case split for (a) ($c = 0$ versus $c \neq 0$), and for (b) it omits the one extra fact the whole proof turns on: a Cauchy sequence is bounded. Since the Cauchy Criterion is banned, boundedness of $\{a_n\}$ must be proved directly from the Cauchy definition (not derived from "Cauchy $\iff$ convergent, convergent $\implies$ bounded"), and that proof is supplied below.

Method 1: definition-only Cauchy estimates

(a) $\{ca_n\}$ is Cauchy.

Idea. (*Scratch work – not part of the written answer.*) If $c = 0$ the claim is trivial ($ca_n = 0$ always). If $c \neq 0$, mimic the Algebraic Limit Theorem's scalar-multiple proof but with a_n, a_m in place of a_n, a : make $|a_n - a_m| < \varepsilon/|c|$ and multiply through by $|c|$.

Proof. Case $c = 0$. Then $ca_n = 0$ for every $n \in \mathbb{N}$, so $|ca_n - ca_m| = 0 < \varepsilon$ for every $\varepsilon > 0$ and all $n, m \in \mathbb{N}$; hence $\{ca_n\}$ is Cauchy.

Case $c \neq 0$. Let $\varepsilon > 0$ be given. Since $\{a_n\}$ is Cauchy, by the definition of Cauchy sequence there is $N \in \mathbb{N}$ such that for all $n, m \geq N$, $|a_n - a_m| < \varepsilon/|c|$. Then for all $n, m \geq N$,

$$|ca_n - ca_m| = |c| |a_n - a_m| < |c| \cdot \frac{\varepsilon}{|c|} = \varepsilon.$$

Therefore, in either case, by the definition of Cauchy sequence, $\{ca_n\}$ is Cauchy. $\square$

$$\boxed{\{ca_n\} \text{ is Cauchy, for every } c \in \mathbb{R}.}$$

(b) $\{a_n^2\}$ is Cauchy; also the sample midterm Q4 model answer.

Idea. (*Scratch work – not part of the written answer.*) $a_n^2 - a_m^2 = (a_n - a_m)(a_n + a_m)$, so $|a_n^2 - a_m^2| \leq |a_n - a_m| \cdot (|a_n| + |a_m|)$. If $\{a_n\}$ is bounded by some M , this is at most $2M|a_n - a_m|$, which can be made small once $|a_n - a_m|$ is small. The missing ingredient is boundedness of a Cauchy sequence, proved directly (Cauchy Criterion is banned, so we cannot go via "Cauchy $\implies$ convergent $\implies$ bounded").

Proof. *Step 1: the definition of Cauchy sequence.* A sequence $\{a_n\}$ is said to be Cauchy if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $n, m \in \mathbb{N}$ with $n, m \geq N$, $|a_n - a_m| < \varepsilon$.

Step 2: every Cauchy sequence is bounded. We show this directly from the definition above. Since $\{a_n\}$ is Cauchy, taking $\varepsilon = 1 > 0$ gives some $N_0 \in \mathbb{N}$ such that for all $n, m \geq N_0$, $|a_n - a_m| < 1$. In particular, taking $m = N_0$, for all $n \geq N_0$,

$$|a_n| = |(a_n - a_{N_0}) + a_{N_0}| \leq |a_n - a_{N_0}| + |a_{N_0}| < 1 + |a_{N_0}|.$$

So every term of the tail $a_{N_0}, a_{N_0+1}, \dots$ is bounded by $1 + |a_{N_0}|$. The remaining, finitely many terms $a_1, \dots, a_{N_0-1}$ each have a finite absolute value, so

$$M := \max\{|a_1|, \dots, |a_{N_0-1}|, 1 + |a_{N_0}|\}$$

is a finite positive number with $|a_n| \leq M$ for every $n \in \mathbb{N}$. Therefore $\{a_n\}$ is bounded.

Step 3: the main estimate. Since $\{a_n\}$ is bounded (Step 2), fix $M > 0$ with $|a_n| \leq M$ for all $n \in \mathbb{N}$. Let $\varepsilon > 0$ be given. Since $\{a_n\}$ is Cauchy, there is $N \in \mathbb{N}$ such that for all $n, m \geq N$, $|a_n - a_m| < \frac{\varepsilon}{2M}$. Then for all $n, m \geq N$,

$$\begin{aligned} |a_n^2 - a_m^2| &= |(a_n - a_m)(a_n + a_m)| = |a_n - a_m| \cdot |a_n + a_m| \\ &\leq |a_n - a_m| (|a_n| + |a_m|) \leq |a_n - a_m| \cdot 2M < \frac{\varepsilon}{2M} \cdot 2M = \varepsilon. \end{aligned}$$

Therefore, by the definition of Cauchy sequence, $\{a_n^2\}$ is Cauchy. □

$\{a_n^2\}$ is Cauchy (boundedness of $\{a_n\}$, proved directly from the Cauchy definition, is the key extra ingredient).

Common slip. Quoting "every Cauchy sequence converges, and convergent sequences are bounded" to get boundedness in part (b) silently smuggles in the Cauchy Criterion, which the question explicitly forbids; boundedness must instead be proved from the raw $\varepsilon = 1$ argument in Step 2 above (which is exactly how the Theorem (Boundedness of Convergent Sequences) is proved in Lecture 4, adapted from "eventually within 1 of the limit" to "eventually within 1 of a_{N_0} "). A second common slip is bounding $|a_n + a_m|$ by $2M$ using the *same* M that bounds $|a_n|$ and $|a_m|$ separately without justifying $|a_n| + |a_m| \leq 2M$ explicitly.

Question A1 (A linear recurrence $b_n = a_{n+1} + ra_n$ forces $\{a_n\}$ to converge)

Problem

Prove that for any sequence $\{a_n\}$ and any real number r with $|r| < 1$, if the sequence $\{b_n := a_{n+1} + ra_n\}$ is convergent, then the sequence $\{a_n\}$ is convergent. (*Tutorial 2, Additional Question A1.*)

Solution

Official hint. (i) Use the Algebraic Limit Theorem to determine the limit a of $\{a_n\}$, if it converges. (ii) Use the definition of the limit of $\{b_n\}$ to deduce the definition of the limit of $\{a_n\}$. (iii) Get an upper bound on $|a_{n+1} - a|$ in terms of $|a_n - a|$; prove inductively an upper bound on $|a_{n+k} - a|$ in terms of $|a_n - a|$ for all $k \in \mathbb{N}$; expect a geometric progression with ratio r .

Beyond the hints. The hint does not derive the target value $a = b/(1+r)$, does not carry out the induction on $|a_{n+k} - a|$, and does not assemble the final ε - N argument combining geometric decay with $b_n \rightarrow b$. All three are supplied below, together with a short standalone lemma that $|r|^k \rightarrow 0$ (needed for the geometric-decay step, and not otherwise available since the Algebraic Limit Theorem alone does not give convergence rates).

Method 1: guess the limit via the Algebraic Limit Theorem, then prove it by an inductive geometric-decay estimate

Idea. (*Scratch work – not part of the written answer.*) *Finding the target limit.* If $\{a_n\}$ converges to some a , then $\{a_{n+1}\}$ (a tail of $\{a_n\}$) also converges to a , so by the Algebraic Limit Theorem $b_n = a_{n+1} + ra_n \rightarrow a + ra = (1+r)a$. Since $|r| < 1$ gives $r > -1$, i.e. $1+r > 0$, the candidate limit is $a := b/(1+r)$, where $b := \lim b_n$.

Setting up the recursion. Let $e_n := a_n - a$. Then

$$b_n - b = (a_{n+1} + ra_n) - (1+r)a = (a_{n+1} - a) + r(a_n - a) = e_{n+1} + re_n,$$

so $e_{n+1} = (b_n - b) - re_n$, hence $|e_{n+1}| \leq |b_n - b| + |r||e_n|$.

Geometric-decay lemma. If $|r| < 1$, then $|r|^k \rightarrow 0$ as $k \rightarrow \infty$. Proof: if $r = 0$ this is trivial ($r^k = 0$ for $k \geq 1$). If $r \neq 0$, write $|r| = 1/(1+h)$ for $h := 1/|r| - 1 > 0$ (using $0 < |r| < 1$). By Bernoulli's inequality, $(1+h)^k \geq 1+kh$ for all $k \in \mathbb{N}$, so $|r|^k = 1/(1+h)^k \leq 1/(1+kh)$. Given $\varepsilon > 0$, the Archimedean Property gives $K \in \mathbb{N}$ with $K > 1/(h\varepsilon)$, so $1/(1+Kh) < 1/(Kh) < \varepsilon$; hence for $k \geq K$, $|r|^k \leq 1/(1+kh) \leq 1/(1+Kh) < \varepsilon$. This proves $|r|^k \rightarrow 0$.

Proof. *Step 1: the target limit.* Let $b := \lim b_n$ (exists by hypothesis) and set $a := \frac{b}{1+r}$; this is well defined since $|r| < 1$ implies $1+r > 0$.

Step 2: a geometric bound on $|e_{n+k}|$. Let $e_n := a_n - a$ for each $n \in \mathbb{N}$. As shown in the Idea, $e_{n+1} = (b_n - b) - re_n$ for every $n \in \mathbb{N}$, so $|e_{n+1}| \leq |r||e_n| + |b_n - b|$.

Fix $\eta > 0$. Since $b = \lim b_n$, there is $N_0 \in \mathbb{N}$ such that for all $n \geq N_0$, $|b_n - b| < \eta$. We prove by induction on $k \geq 0$ that for every $k \in \mathbb{N}_0$,

$$|e_{N_0+k}| \leq |r|^k |e_{N_0}| + \eta \sum_{j=0}^{k-1} |r|^j$$

(with the empty sum, for $k = 0$, understood as 0). The base case $k = 0$ is the equality $|e_{N_0}| \leq |e_{N_0}|$. For the inductive step, suppose the bound holds for some $k \geq 0$. Since $N_0 + k \geq N_0$, we have $|b_{N_0+k} - b| < \eta$, so

$$\begin{aligned} |e_{N_0+k+1}| &\leq |r| |e_{N_0+k}| + |b_{N_0+k} - b| \\ &< |r| \left(|r|^k |e_{N_0}| + \eta \sum_{j=0}^{k-1} |r|^j \right) + \eta \\ &= |r|^{k+1} |e_{N_0}| + \eta \sum_{j=1}^k |r|^j + \eta \\ &= |r|^{k+1} |e_{N_0}| + \eta \sum_{j=0}^k |r|^j, \end{aligned}$$

which is the claimed bound for $k + 1$. This completes the induction. Since $0 \leq |r| < 1$, the finite geometric identity $(1 - |r|) \sum_{j=0}^{k-1} |r|^j = 1 - |r|^k \leq 1$ gives $\sum_{j=0}^{k-1} |r|^j \leq \frac{1}{1 - |r|}$, so for every $k \in \mathbb{N}_0$,

$$|e_{N_0+k}| \leq |r|^k |e_{N_0}| + \frac{\eta}{1 - |r|}.$$

Step 3: the ε -N assembly. Let $\varepsilon > 0$ be given. Apply Step 2 with $\eta := \frac{\varepsilon(1 - |r|)}{2} > 0$, obtaining $N_0 \in \mathbb{N}$ and the bound $|e_{N_0+k}| \leq |r|^k |e_{N_0}| + \varepsilon/2$ for every $k \in \mathbb{N}_0$. If $e_{N_0} = 0$, take $K = 0$; otherwise, by the geometric-decay lemma, $|r|^k \rightarrow 0$, so there is $K \in \mathbb{N}$ such that for all $k \geq K$, $|r|^k |e_{N_0}| < \varepsilon/2$. Set $N := N_0 + K$. For every $n \geq N$, write $n = N_0 + k$ with $k = n - N_0 \geq K$; since $|r| < 1$ implies $|r|^k \leq |r|^K$ for $k \geq K$,

$$|a_n - a| = |e_n| = |e_{N_0+k}| \leq |r|^k |e_{N_0}| + \frac{\varepsilon}{2} \leq |r|^K |e_{N_0}| + \frac{\varepsilon}{2} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Therefore, by the definition of the limit of a sequence, $a = \lim a_n$; in particular $\{a_n\}$ converges. $\square$

$\{a_n\} \text{ converges, with } \lim a_n = \frac{\lim b_n}{1 + r}.$

Common slip. Assuming from the start that $\{a_n\}$ converges (to invoke the Algebraic Limit Theorem and read off $a = b/(1 + r)$) and stopping there is circular: the Algebraic Limit Theorem step only identifies the *candidate* limit; it does not establish convergence. The actual proof of convergence must instead be the ε -N argument of Steps 2–3, entirely independent of assuming $\{a_n\}$ converges.

Question A2 (Cesaro means of a convergent sequence)

Problem

Show that if the sequence $\{a_n\}$ has the limit a , then the sequence $\{b_n\}$ with

$$b_n = \frac{a_1 + a_2 + \cdots + a_n}{n}$$

for every $n \in \mathbb{N}$ converges to a . (*Tutorial 2, Additional Question A2.*)

Solution

Official hint. (i) Express the distance between b_n and a in terms of the distances between every a_k and a for $k \in \{1, \dots, n\}$. (ii) Use the triangle inequality. (iii) Recall that convergent sequences are bounded. (iv) The Archimedean Property can be useful.

Beyond the hints. The hint does not show the split of the averaged sum into a bounded initial block (whose average vanishes as $n \rightarrow \infty$) and a tail block (each term within ε of a , so its average is within ε of a); this two-part estimate, and the resulting choice of N , is supplied below.

Method 1: split the average into a fixed head and a controlled tail

Idea. (*Scratch work – not part of the written answer.*)

$$b_n - a = \frac{1}{n} \sum_{k=1}^n a_k - a = \frac{1}{n} \sum_{k=1}^n (a_k - a).$$

Fix a block size M : split $\sum_{k=1}^n (a_k - a) = \sum_{k=1}^M (a_k - a) + \sum_{k=M+1}^n (a_k - a)$. The first block has a fixed number of terms, each bounded (Boundedness of Convergent Sequences), so it contributes a fixed constant divided by n , which vanishes as $n \rightarrow \infty$. The second block, once M is chosen so that a_k is within $\varepsilon/2$ of a for all $k > M$, contributes at most $\varepsilon/2$ to the average.

Proof. Since $\{a_n\}$ converges, by the Theorem (Boundedness of Convergent Sequences) there is $C > 0$ with $|a_n| \leq C$ for every $n \in \mathbb{N}$; consequently $|a_n - a| \leq |a_n| + |a| \leq C + |a| =: C'$ for every $n \in \mathbb{N}$.

Let $\varepsilon > 0$ be given. Since $a = \lim a_n$, there is $M \in \mathbb{N}$ such that for all $k > M$, $|a_k - a| < \frac{\varepsilon}{2}$. By the Archimedean Property, choose $N \in \mathbb{N}$ with $N > M$ and $N > \frac{2MC'}{\varepsilon}$.

Fix $n \geq N$ (so in particular $n > M$). Then

$$b_n - a = \frac{1}{n} \sum_{k=1}^n (a_k - a) = \frac{1}{n} \sum_{k=1}^M (a_k - a) + \frac{1}{n} \sum_{k=M+1}^n (a_k - a).$$

By the triangle inequality, and since each of the M terms in the first sum satisfies $|a_k - a| \leq C'$,

$$\left| \frac{1}{n} \sum_{k=1}^M (a_k - a) \right| \leq \frac{1}{n} \sum_{k=1}^M |a_k - a| \leq \frac{MC'}{n}.$$

Since there are $n - M \leq n$ terms in the second sum, each with $|a_k - a| < \varepsilon/2$,

$$\left| \frac{1}{n} \sum_{k=M+1}^n (a_k - a) \right| \leq \frac{1}{n} \sum_{k=M+1}^n |a_k - a| < \frac{1}{n} \cdot (n - M) \cdot \frac{\varepsilon}{2} \leq \frac{\varepsilon}{2}.$$

Combining the two bounds by the triangle inequality,

$$|b_n - a| \leq \frac{MC'}{n} + \frac{\varepsilon}{2}.$$

Since $n \geq N > \frac{2MC'}{\varepsilon}$, we have $\frac{MC'}{n} < \frac{\varepsilon}{2}$. Therefore $|b_n - a| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$. Since $n \geq N$ was arbitrary, we conclude, by the definition of the limit of a sequence, that $a = \lim b_n$. $\square$

$$a = \lim b_n, \quad \text{where } b_n = \frac{a_1 + \cdots + a_n}{n}.$$

Common slip. Trying to bound $|b_n - a|$ by a single application of the triangle inequality across all n terms at once, using only the tail bound $|a_k - a| < \varepsilon/2$, silently assumes that bound holds for *every* $k \in \{1, \dots, n\}$, including the small ones – but the definition of $a = \lim a_n$ only guarantees it for $k > M$. The finitely many "bad" early terms $a_1, \dots, a_M$ must be handled separately, and it is exactly their fixed count (not growing with n) that lets $MC'/n \rightarrow 0$.

Question A3 (Doubly indexed arrays: iterated limits versus the double limit)

Problem

Given a doubly indexed array a_{mn} where $m, n \in \mathbb{N}$, define $a = \lim_{m,n \rightarrow \infty} a_{mn}$ to mean

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall m, n \geq N \quad |a_{mn} - a| < \varepsilon.$$

[Note: a doubly indexed array is a function $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ where we use a_{mn} to denote $f(m, n)$.]

- If the array satisfies $a_{mn} = m/(m+n)$ for each $m, n \in \mathbb{N}$, what are the iterated limits $\lim_{n \rightarrow \infty} (\lim_{m \rightarrow \infty} a_{mn})$ and $\lim_{m \rightarrow \infty} (\lim_{n \rightarrow \infty} a_{mn})$?
- If the array satisfies $a_{mn} = mn/(m+n)$ for each $m, n \in \mathbb{N}$, does there exist $a \in \mathbb{R}$ for which $a = \lim_{m,n \rightarrow \infty} a_{mn}$? Do the two iterated limits exist? How do these three values compare?
- Prove that if $a = \lim_{m,n \rightarrow \infty} a_{mn}$ and the iterated limit $\lim_{n \rightarrow \infty} (\lim_{m \rightarrow \infty} a_{mn})$ exists, then the iterated limit is a .

(Tutorial 2, Additional Question A3.)

Solution

Official hint. (a) For each $n \in \mathbb{N}$, the sequence $\{a_{mn}\}_{m=1}^{\infty}$ has a limit denoted b_n ; the iterated limit $\lim_{n \rightarrow \infty} (\lim_{m \rightarrow \infty} a_{mn})$ is $\lim_n b_n$; the other iterated limit swaps the roles of m, n . (b) "No. When $m = n$, $a_{mn} = \frac{1}{2}$; but when $m = 2n$, $a_{mn} = \frac{2}{5} \neq \frac{1}{2}$. Yes, $0 = \lim_{n \rightarrow \infty} (\lim_{m \rightarrow \infty} mn/(m^2 + n^2))$ and $0 = \lim_{m \rightarrow \infty} (\lim_{n \rightarrow \infty} mn/(m^2 + n^2))$." (c) No hint beyond the notation of part (a).

Note on the official hint. The question sheet prints part (b) with $a_{mn} = mn/(m+n)$ (confirmed against the rendered question PDF). Checking the hint's own numbers shows this is *not* the formula the hint actually uses: at $m = n$, $mn/(m+n) = n^2/(2n) = n/2$, which is not the fixed value $\frac{1}{2}$ the hint states; but $mn/(m^2+n^2)$ at $m = n$ gives $n^2/(2n^2) = \frac{1}{2}$, and at $m = 2n$ gives $2n^2/(5n^2) = \frac{2}{5}$ – exactly the hint's two numbers. So the entire hint for part (b), not merely its last sentence, is computed for $a_{mn} = mn/(m^2+n^2)$, while the question as printed asks about $a_{mn} = mn/(m+n)$. Below, part (b) is solved in full for the printed formula $mn/(m+n)$ first, since that is what the question asks; the hint's formula $mn/(m^2+n^2)$ is then solved separately to reconcile the two.

Method 1: reduce each iterated limit to a single-variable ε - N computation

(a) $a_{mn} = m/(m+n)$.

Idea. (Scratch work – not part of the written answer.) For fixed n , as $m \rightarrow \infty$ the ratio $m/(m+n) \rightarrow 1$ (the n becomes negligible next to m). For fixed m , as $n \rightarrow \infty$ the ratio $m/(m+n) \rightarrow 0$ (the denominator grows without bound while the numerator is fixed).

Proof. *Inner limit as $m \rightarrow \infty$, n fixed.* Fix $n \in \mathbb{N}$. For every $m \in \mathbb{N}$,

$$\left| \frac{m}{m+n} - 1 \right| = \left| \frac{m - (m+n)}{m+n} \right| = \frac{n}{m+n} < \frac{n}{m}.$$

Given $\varepsilon > 0$, the Archimedean Property gives $M \in \mathbb{N}$ with $M > n/\varepsilon$; then for $m \geq M$, $|m/(m+n) - 1| < n/m \leq n/M < \varepsilon$. Hence $\lim_{m \rightarrow \infty} a_{mn} = 1$ for every $n \in \mathbb{N}$; write $b_n := 1$. Since $\{b_n\}$ is the constant sequence 1, $\lim_{n \rightarrow \infty} b_n = 1$. Therefore

$$\lim_{n \rightarrow \infty} \left(\lim_{m \rightarrow \infty} a_{mn} \right) = 1.$$

Inner limit as $n \rightarrow \infty$, m fixed. Fix $m \in \mathbb{N}$. For every $n \in \mathbb{N}$,

$$\left| \frac{m}{m+n} - 0 \right| = \frac{m}{m+n} < \frac{m}{n}.$$

Given $\varepsilon > 0$, the Archimedean Property gives $N \in \mathbb{N}$ with $N > m/\varepsilon$; then for $n \geq N$, $|m/(m+n)| < m/n \leq m/N < \varepsilon$. Hence $\lim_{n \rightarrow \infty} a_{mn} = 0$ for every $m \in \mathbb{N}$; write $c_m := 0$. Since $\{c_m\}$ is the constant sequence 0, $\lim_{m \rightarrow \infty} c_m = 0$. Therefore

$$\lim_{m \rightarrow \infty} \left(\lim_{n \rightarrow \infty} a_{mn} \right) = 0.$$

□

(b), as printed: $a_{mn} = mn/(m+n)$.

Idea. (*Scratch work – not part of the written answer.*) Along the diagonal $m = n$, $a_{nn} = n^2/(2n) = n/2 \rightarrow \infty$; this single unbounded path already rules out a finite double limit. The same growth shows up in each iterated limit.

Proof: the double limit does not exist. Suppose, for contradiction, that $a = \lim_{m,n \rightarrow \infty} a_{mn}$ for some $a \in \mathbb{R}$. Taking $\varepsilon = 1$ in the definition, there is $N \in \mathbb{N}$ such that for all $m, n \geq N$, $|a_{mn} - a| < 1$, so in particular $a_{mn} < a + 1$ for all $m, n \geq N$. By the Archimedean Property, choose $n_0 \in \mathbb{N}$ with $n_0 \geq N$ and $n_0 > 2(a + 1)$. Then $m = n = n_0 \geq N$, but

$$a_{n_0 n_0} = \frac{n_0 \cdot n_0}{n_0 + n_0} = \frac{n_0}{2} > a + 1,$$

contradicting $a_{n_0 n_0} < a + 1$. Hence no such a exists; the double limit $\lim_{m,n \rightarrow \infty} a_{mn}$ does not exist (a_{mn} diverges to $+\infty$ along the diagonal). □

Proof: neither iterated limit exists. Fix $n \in \mathbb{N}$. For every $m \in \mathbb{N}$,

$$\left| \frac{mn}{m+n} - n \right| = \left| \frac{mn - n(m+n)}{m+n} \right| = \frac{n^2}{m+n} < \frac{n^2}{m}.$$

Given $\varepsilon > 0$, the Archimedean Property gives $M > n^2/\varepsilon$; then $m \geq M$ gives $|a_{mn} - n| < \varepsilon$. Hence $\lim_{m \rightarrow \infty} a_{mn} = n$ for every fixed $n \in \mathbb{N}$; write $b_n := n$. But $\{b_n\} = \{n\}$ is unbounded (Archimedean Property, as in Q1(d)), so by the Theorem (Boundedness of Convergent Sequences) $\{b_n\}$ cannot converge. Hence $\lim_{n \rightarrow \infty} (\lim_{m \rightarrow \infty} a_{mn}) = \lim_{n \rightarrow \infty} n$ does not exist.

By the symmetry $a_{mn} = a_{nm}$ of the formula $mn/(m+n)$ in m and n , the same computation with the roles of m, n exchanged gives $\lim_{n \rightarrow \infty} a_{mn} = m$ for every fixed $m \in \mathbb{N}$, so $\lim_{m \rightarrow \infty} m$ also does not exist.

Comparison. For the formula as printed, all three quantities – the double limit and both iterated limits – fail to exist, and all three “diverge to $+\infty$ ” in the same qualitative sense (unlike part (a), where the double limit’s absence still left two well-defined but *different* finite iterated limits, 1 and 0).

As printed ($a_{mn} = mn/(m+n)$): the double limit and both iterated limits all fail to exist.

□

Note on the official hint: solving the hint’s actual formula $a_{mn} = mn/(m^2 + n^2)$.

Proof: the double limit does not exist. Along $m = n$, $a_{nn} = n^2/(2n^2) = \frac{1}{2}$ for every $n \in \mathbb{N}$; along $m = 2n$, $a_{2n,n} = 2n^2/(4n^2 + n^2) = \frac{2}{5}$ for every $n \in \mathbb{N}$. Suppose, for contradiction, $a = \lim_{m,n \rightarrow \infty} a_{mn}$ for some $a \in \mathbb{R}$. Take $\varepsilon := \frac{1}{20}$ (note $|\frac{1}{2} - \frac{2}{5}| = \frac{1}{10} = 2\varepsilon$); there is $N \in \mathbb{N}$ such that for all $m, n \geq N$, $|a_{mn} - a| < \varepsilon$. Both $(m, n) = (N, N)$ and $(m, n) = (2N, N)$ satisfy $m, n \geq N$, so $|\frac{1}{2} - a| < \varepsilon$ and $|\frac{2}{5} - a| < \varepsilon$. By the triangle inequality,

$$\frac{1}{10} = \left| \frac{1}{2} - \frac{2}{5} \right| \leq \left| \frac{1}{2} - a \right| + \left| a - \frac{2}{5} \right| < \varepsilon + \varepsilon = \frac{1}{10},$$

i.e. $\frac{1}{10} < \frac{1}{10}$, a contradiction. Hence the double limit does not exist. □

Proof: both iterated limits equal 0. Fix $n \in \mathbb{N}$. For every $m \in \mathbb{N}$, since $m^2 + n^2 \geq m^2$,

$$\left| \frac{mn}{m^2 + n^2} \right| \leq \frac{mn}{m^2} = \frac{n}{m}.$$

Given $\varepsilon > 0$, choose $M > n/\varepsilon$ (Archimedean Property); then $m \geq M$ gives $|a_{mn}| < \varepsilon$. Hence $\lim_{m \rightarrow \infty} a_{mn} = 0$ for every fixed n , so $\lim_{n \rightarrow \infty} (\lim_{m \rightarrow \infty} a_{mn}) = \lim_n 0 = 0$. By the symmetry of $mn/(m^2 + n^2)$ in m, n , the same argument gives $\lim_{n \rightarrow \infty} a_{mn} = 0$ for every fixed m , so $\lim_{m \rightarrow \infty} (\lim_{n \rightarrow \infty} a_{mn}) = 0$ as well. This matches the hint’s “Yes, $0 = \dots$ ” line for both iterated limits.

Hint’s formula ($a_{mn} = mn/(m^2 + n^2)$): double limit does not exist; both iterated limits equal 0.

□

(c).

Idea. (*Scratch work – not part of the written answer.*) Write $b_n := \lim_{m \rightarrow \infty} a_{mn}$ (assumed to exist for each n , as part of “the iterated limit exists”), and let $L := \lim_{n \rightarrow \infty} b_n$ (also assumed to exist). We must show $L = a$. The cleanest route avoids citing the (partly unproved) Order Limit Theorem: fix n large, use the double-limit hypothesis to control a_{mn} for all large m simultaneously, and let $m \rightarrow \infty$ by a direct triangle-inequality argument, rather than by formally “taking a limit inside an inequality”.

Proof. Let $\varepsilon > 0$ be given. Since $a = \lim_{m,n \rightarrow \infty} a_{mn}$, there is $N_1 \in \mathbb{N}$ such that for all $m, n \geq N_1$,

$$|a_{mn} - a| < \frac{\varepsilon}{2}.$$

Fix any $n \geq N_1$. We claim $|b_n - a| \leq \varepsilon/2$, where $b_n = \lim_{m \rightarrow \infty} a_{mn}$. Suppose, for contradiction, that $|b_n - a| > \varepsilon/2$; let $\delta := |b_n - a| - \varepsilon/2 > 0$. Since $b_n = \lim_{m \rightarrow \infty} a_{mn}$, there is $m_0 \geq N_1$ such that $|a_{m_0 n} - b_n| < \delta$. Then, by the triangle inequality,

$$|b_n - a| \leq |b_n - a_{m_0 n}| + |a_{m_0 n} - a| < \delta + \frac{\varepsilon}{2} = \left(|b_n - a| - \frac{\varepsilon}{2}\right) + \frac{\varepsilon}{2} = |b_n - a|,$$

i.e. $|b_n - a| < |b_n - a|$, a contradiction. Hence $|b_n - a| \leq \varepsilon/2$.

Since $n \geq N_1$ was an arbitrary index with $n \geq N_1$, this shows $|b_n - a| \leq \varepsilon/2 < \varepsilon$ for every $n \geq N_1$. As $\varepsilon > 0$ was arbitrary, this is precisely the statement that $a = \lim_{n \rightarrow \infty} b_n$.

Finally, by hypothesis $L = \lim_{n \rightarrow \infty} b_n$ also exists. By the Theorem (Uniqueness of Limits), a sequence has at most one limit, so $L = a$. $\square$

If $a = \lim_{m, n \rightarrow \infty} a_{mn}$ and $\lim_{n \rightarrow \infty} (\lim_{m \rightarrow \infty} a_{mn})$ exists, it equals a .

Common slip. In part (b), silently using the hint's numbers ($\frac{1}{2}$ and $\frac{2}{5}$) while writing the printed formula $mn/(m+n)$ produces an internally inconsistent solution (those numbers simply do not come from that formula); the printed question and the hint must be solved as two separate, clearly labelled computations, as above. In part (c), the frequent shortcut "let $m \rightarrow \infty$ in the inequality $|a_{mn} - a| < \varepsilon/2$ " is informal; the direct triangle-inequality argument given above makes that step rigorous without invoking the unproved parts of the Order Limit Theorem.

Method Map

Q.	Core topic	Methods included
1	8 true/false items on convergence	Explicit examples with ε -based non-convergence checks (True items); proofs by contradiction via the Algebraic Limit Theorem, Boundedness of Convergent Sequences, and the triangle inequality (False items).
2	ε - N verification of three limits	Algebraic bounding of $ a_n - a $ by C/n or $C/n^{1/3}$, then the Archimedean Property to invert for $N(\varepsilon)$.
3	Bounded $\times$ null, and a nonzero-limit warning	Squeeze Theorem sandwich $-M b_n \leq a_n b_n \leq M b_n $; explicit oscillating counterexample $(-1)^n$.
4	Three definition-only proofs (ALT banned)	Direct ε - N bounding: linear rescaling; rationalizing $\sqrt{a_n} - \sqrt{2}$; two-stage ε' -then- ε bound for $1/a_n$.
5	Cauchy sequences: scalar multiples, squares	Case split on $c = 0$; direct proof that Cauchy $\implies$ bounded (no Cauchy Criterion), then $ a_n^2 - a_m^2 = a_n - a_m a_n + a_m \leq 2M a_n - a_m $.
A1	Linear recurrence $b_n = a_{n+1} + r a_n$	Algebraic Limit Theorem to guess the target limit $b/(1+r)$; inductive geometric-decay bound on $ a_{n+k} - a $; Bernoulli's inequality for $ r ^k \rightarrow 0$.
A2	Cesaro means	Split the averaged sum into a fixed bounded head and a controlled $\varepsilon/2$ -tail; Boundedness of Convergent Sequences; Archimedean Property.
A3	Doubly indexed arrays: double vs. iterated limits	Single-variable ε - N reduction for each inner limit; direct triangle-inequality argument (avoiding unproved Order Limit Theorem parts) for part (c).

Technique Summary

1. **"Infinitely many 1's" forces the limit to be 1, not just close to it.** A single specific term equal to 1, occurring past the tail index N , collides with an arbitrarily small ε -ball around any other candidate limit.
2. **Enlarge before you invert.** To find $N(\varepsilon)$, first bound $|a_n - a|$ by a simpler expression C/n^p that is monotone in n , then solve $C/n^p < \varepsilon$ for n ; the resulting N need not be sharp.
3. **The Squeeze Theorem needs matching outer limits, not just an inequality.** $|a_n| \leq M$ alone only gives $-M|b_n| \leq a_n b_n \leq M|b_n|$; it is $b_n \rightarrow 0$ that forces both outer sequences to the same limit.
4. **Rationalizing beats a two-sided ε' -interval when positivity is already given.** For $\sqrt{a_n} - \sqrt{2}$, multiplying by the conjugate gives a one-line bound once $a_n > 0$ is assumed outright.
5. **Reciprocals need a floor first.** Bounding $1/a_n$ requires an initial application of the limit definition (with a throwaway $\varepsilon' = 1$) to push a_n away from 0, before the main ε -estimate.

6. **A Cauchy sequence is bounded – prove it from scratch when the Cauchy Criterion is banned.** The same $\varepsilon = 1$ -then-max-of-finitely-many-terms trick that bounds convergent sequences (Lecture 4) works verbatim from the Cauchy definition alone.
7. **Guessing the limit via the Algebraic Limit Theorem is not a proof of convergence.** It only identifies the candidate; an independent ε - N (or geometric-decay/induction) argument is still required.
8. **A double limit existing is strictly stronger than either iterated limit existing.** All three values can disagree, one can exist without the others, or (as in A3(b) printed) all three can simultaneously fail to exist.

Final Answers

1. (a) True, (b) False, (c) True, (d) True, (e) False, (f) True, (g) False, (h) True.
2. All three limits verified directly from the ε - N definition; $N(\varepsilon) = \lceil 3/(25\varepsilon) \rceil$, $\lceil 2/\varepsilon \rceil$, $\lceil 1/\varepsilon^3 \rceil$ respectively (any sufficiently large integer works).
3. (a) $0 = \lim a_n b_n$, by the Squeeze Theorem. (b) $a_n = (-1)^n$, $b_n = 1$ gives $a_n b_n = (-1)^n$, divergent.
4. (a) $1 = \lim \frac{2a_n - 1}{3}$. (b) $\sqrt{2} = \lim \sqrt{a_n}$. (c) $\frac{1}{2} = \lim \frac{1}{a_n}$.
5. (a) $\{ca_n\}$ is Cauchy for every $c \in \mathbb{R}$. (b) $\{a_n^2\}$ is Cauchy (using boundedness of Cauchy sequences, proved directly).
6. $\{a_n\}$ converges, with $\lim a_n = (\lim b_n)/(1 + r)$.
7. $a = \lim b_n$, where $b_n = (a_1 + \cdots + a_n)/n$.
8. (a) $\lim_n(\lim_m a_{mn}) = 1$, $\lim_m(\lim_n a_{mn}) = 0$. (b) As printed ($mn/(m+n)$): double limit and both iterated limits all fail to exist. Hint's formula ($mn/(m^2 + n^2)$): double limit fails to exist; both iterated limits equal 0. (c) The iterated limit, when it exists, equals the double limit a .